

Nguyen Tong (Toby) Tran

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EDUCATION

University of South Australia

Bachelor of Information Technology (Software Development)

Mawson Lakes, SA

Mar. 2023 - Nov. 2025

- **GPA:** 6.17 / 7.0
- **Honours/Awards:** 2023-25 University Merit Awards
- **Relevant Coursework:** Data Structures Essentials, Object-Oriented Programming (OOP), Design Patterns with C++, Data-Driven Web Technologies, Web Technology, System Requirements and User Experience, Agile Development and Governance, Operating Systems and Tool Chains.

Marryatville High School

- **Honours/Awards:** 2022 SACE Merit Award

Marryatville, SA

2020 - 2022

TECHNICAL SKILLS

Languages: C#, Python, JavaScript, HTML/CSS, MySQL

Frameworks & Library: ASP.NET Core (MVC & WebAPI), FastAPI, ReactJS

Database: SQL Server, EF Core, LINQ, MongoDB

Tools & Methodologies: Git, Jira Atlassian (Scrum-based)

FEATURED PROJECTS

Mission Planner: FastAPI, PyWebView, ReactJS, TailwindCSS, CesiumJS, DBScan Nov. 2025

- Designed a layered backend architecture via domain-specific functionalities and the use of design patterns (*Strategy, Adapter, Factory*) for easier maintenance and future extensions.
- Optimised backend performance by implementing **10+ asynchronous FastAPI routes** for more efficient data processing, alongside **custom logging systems** to facilitate debugging.
- Integrated a *ReactJS* user interface into a **native desktop environment** via *PyWebView* and *Nuitka*, allowing a more engaging user experience than traditional native UIs.
- Implemented *Rechart.js* components to visualise and provide insights from massive datasets.
- Built a **synchroniser** that leverages engineered data scraping models and internal logs to handle automatic bulk data downloads, serialisation, and synchronisation (for up to **100,000+ CSV records**) in real-time while remaining responsive.
- Developed a **3D-globe visualisation section**, powered by *CesiumJS*, to model anticipated conjunctions between **multiple satellites** and space debris.

SA Expiation Dashboard: ASP.NET Core Web API, LINQ, ReactJS, Bootstrap, D3.js Nov. 2024

- Developed a *ReactJS* dashboard with **up to 6 different search/filter controls** to identify **optimal MPDC locations** from expiation data.
- Designed a navigable *Bootstrap-styled* template aligned with SAPOL branding, incorporating **version control** via *Git*.
- Fetched *MS-SQL* data with *LINQ* and *EF Core*, enhanced by *SwaggerGen* for API documentation and testing.
- Visualised MPDC location insights on search results and a report page using **D3.js graphs**.
- Implemented backend in **Model-View-Controller (MVC) pattern** for a clearer separation of concerns.

UTR Web App: Bootstrap, NodeJS/ExpressJS, MongoDB Oct. 2024

- Led UI development for Fuel Listings, Food Menu, Cart, Checkout, and Order Confirmation.
- Implemented page layouts with *Bootstrap* for authentication features.
- Assisted in bug fixes across backend integration.